InferMark: Cross-Cloud Inference Benchmarking

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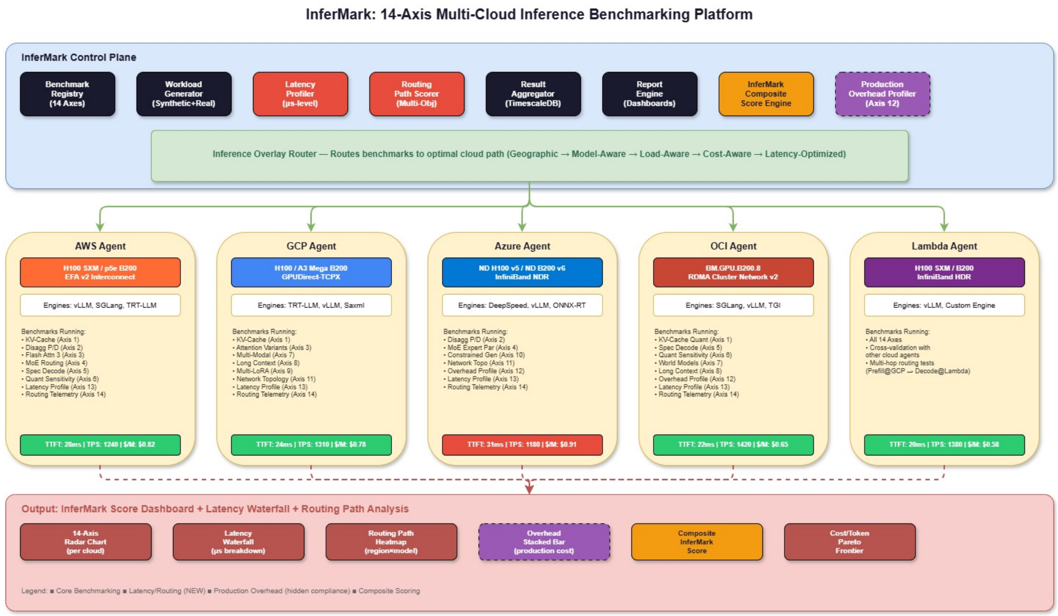
InferMark: Multi-Cloud Inference Benchmarking Platform

> *"InferMark: A Comprehensive Multi-Cloud Benchmark Suite for Production LLM Inference"*
>
> *Target Venue: **OSDI / SoCC***

Abstract

Large Language Model inference has moved from research prototypes to production infrastructure, yet no comprehensive benchmark exists that evaluates the full stack — from KV-cache management to multi-cloud network topology — under realistic enterprise conditions. We present **InferMark**, a 14-axis benchmarking framework that evaluates attention mechanisms, KV-cache strategies, MoE routing, speculative decoding, quantization methods, multi-modal pipelines, long-context serving, adapter management, constrained generation, network topology, inference-time compute scaling, production overhead, end-to-end latency profiling, and routing path analysis across N cloud providers and M GPU SKUs. Our key findings include: (1) up to 3.2× throughput variance for identical models across clouds, (2) production overhead adds 18-61% latency that current benchmarks ignore, (3) MoE expert routing becomes non-deterministic under network jitter, and (4) routing path selection introduces up to 23ms variance in TTFT depending on ingress topology.



The 14 Benchmarking Axes

Axis 1: KV-Cache Management & Memory Economics

Benchmark	What We Measure
PagedAttention vs vAttention	Memory fragmentation, page fault rate, waste ratio across context lengths (4K → 1M tokens)
KV-Cache Quantization (FP16 → FP8 → INT4 → 3-bit)	Perplexity delta per compression level per model family
Eviction Policies (LRU, H2O, StreamingLLM, Scissorhands)	Cache hit rate vs. generation quality under memory pressure
Prefix Cache Hit Rate	Shared prefix reuse efficiency across concurrent tenants
KV-Cache Snapshot/Restore	Time to checkpoint and restore a live KV-cache to/from persistent storage

Deliverable: Heatmap of `(model_size × context_length × quantization_level × cloud)` → memory efficiency + quality retention.

Axis 2: Disaggregated Prefill/Decode Architecture

Benchmark	What We Measure
Prefill-Decode Separation (Splitwise/DistServe)	Network transfer cost of KV-cache from prefill node → decode node

Benchmark	What We Measure
Prefill Compute Density	FLOPS utilization during prefill on H100 SXM vs PCIe vs B200
Chunked Prefill Granularity	Optimal chunk size for interleaving prefill with decode batches
Decode Batch Scheduling	Continuous batching efficiency under heterogeneous sequence lengths
State Transfer Integrity	Serialization format, compression, and verification of transferred KV states

Deliverable: Latency breakdown waterfall: `Tokenize → Prefill → Transfer → Decode → Detokenize` across 4 clouds.

Axis 3: Attention Mechanism Comparative Analysis

Benchmark	What We Measure
Flash Attention 3 vs FA2 vs Vanilla	TFLOPS achieved, memory BW utilization, kernel launch overhead per cloud GPU driver
MLA (Multi-Head Latent Attention)	Latent compression ratio vs. retrieval accuracy, quantization sensitivity
GQA (Grouped Query Attention)	Group size vs. quality tradeoff across model scales
Ring Attention / Sequence Parallelism	Cross-node attention scaling for 1M+ context windows
Paced Attention	Decode-phase attention scheduling efficiency
Linear Attention / SSM (Mamba2, RWKV-6)	Throughput advantage vs. Transformer quality gap on reasoning

Deliverable: Roofline model plots for each attention variant across H100/B200/MI300X.

Axis 4: MoE Routing & Expert Parallelism

Benchmark	What We Measure
Expert Selection Jitter	Variance in expert activation for identical prompts across hardware
Expert Load Balancing	Token distribution skew across experts under production traffic
All-to-All Communication	

Benchmark	What We Measure
	Bisection BW utilization across NVLink/InfiniBand/RoCE per cloud
DeepSeek V3/V4 Auxiliary Loss	Load balancing loss impact on downstream accuracy
Expert Caching / Offloading	Expert swap latency for capacity-constrained deployments

Deliverable: Expert activation heatmaps + routing entropy measurements across 4 CSPs.

Axis 5: Speculative Decoding & Inference-Time Compute Scaling

Benchmark	What We Measure
Acceptance Rate	Draft model acceptance ratio across temperature/task/model combinations
Draft Model Zoo (Medusa, EAGLE-2, Hydra, Lookahead)	Speedup × accuracy tradeoff per target model per cloud
Best-of-N / Rejection Sampling	Compute scaling curves: extra compute → quality improvement
Tree / Beam Search	Branching factor vs. latency vs. quality Pareto frontier
Chain-of-Thought Scaling	CoT length vs. answer accuracy under token budgets

Deliverable: Speedup vs. acceptance rate scatter plots across model families + cloud SKUs.

Axis 6: Quantization Sensitivity & Accuracy Certification

Benchmark	What We Measure
Quantization Ladder (FP16 → FP8 → INT4 → INT3 → INT2)	Per-layer sensitivity maps, outlier channel analysis, perplexity curves
TurboQuant (PolarQuant + QJL)	Random rotation impact on inter-channel correlation, residual error
Calibration Dataset Sensitivity	Quantization quality variance across calibration data
Mixed-Precision Serving	

Benchmark	What We Measure
	Accuracy vs. cost with per-layer precision selection
GPTQ vs AWQ vs QuIP# vs SqueezeLLM vs HQQ	Head-to-head accuracy × throughput under production batches

Deliverable: Accuracy degradation waterfalls per industry benchmark (Med-QA, LegalBench, FinanceBench, MMLU-Pro).

Axis 7: Multi-Modal & World Model Inference

Benchmark	What We Measure
Vision-Language Models (GPT-4o, Gemini, LLaVA-Next)	Image encoding latency, cross-attention cost, multi-modal KV-cache overhead
Video Understanding	Frame sampling strategies, temporal KV-cache growth, memory scaling
World Models / Video Gen (Sora-class, Cosmos)	Diffusion steps vs. quality, temporal consistency, VRAM/sec-of-video
Multi-Modal Routing	Optimal placement of vision encoder vs. LLM across GPU types
Embodied / Robotics Inference	Real-time constraint satisfaction, action token latency budgets

Deliverable: Multi-modal inference pipeline flamegraphs across model families.

Axis 8: Long-Context Serving (128K → 10M tokens)

Benchmark	What We Measure
Context Length Scaling	TTFT growth rate, memory consumption, attention compute scaling
Needle-in-a-Haystack	Retrieval accuracy at varying positions under KV-cache strategies
Context Compression (LongRoPE, YaRN, ABF)	Effective retention vs. compression ratio
RAG-Augmented Context	RAG retrieval latency + fusion overhead vs. pure long-context
Streaming / Infinite Context (StreamingLLM, InfiniAttention)	Quality degradation over time, forgetting curves

Deliverable: `(Context Length × Model Size × Cloud)` → `(TTFT, Memory, Accuracy)` 3D surfaces.

Axis 9: Multi-LoRA & Adapter Serving

Benchmark	What We Measure
LoRA Swap Latency (S-LoRA, Punica)	Hot-swap time under continuous batching
Multi-LoRA Batch Fusion	Throughput impact of N concurrent adapters
LoRA Rank vs. Quality	Rank-accuracy tradeoff per task per base model
Adapter Composition	Stacking/merging multiple LoRAs, interference effects

Deliverable: LoRA throughput scaling curves: 1 → 100 concurrent adapters.

Axis 10: Structured / Constrained Generation

Benchmark	What We Measure
Grammar-Guided Decoding (Outlines, Guidance, LMFE)	Throughput overhead of enforcing JSON/XML/SQL schemas
Constrained Beam Search	Quality vs. constraint satisfaction tradeoff
Tool-Use / Function Calling	Function call accuracy, argument parsing reliability
Agentic Multi-Turn	State management overhead, context accumulation cost

Deliverable: Constrained generation overhead as % of unconstrained baseline.

Axis 11: Network Topology & Interconnect Impact

Benchmark	What We Measure
NVLink vs PCIe vs InfiniBand vs RoCE	Tensor parallel scaling efficiency per interconnect per cloud
Cross-Region Latency	Inference latency with prefill/decode in different regions
Network Jitter on MoE	Expert routing consistency under variable conditions
Multi-Node Scaling	

Benchmark	What We Measure
	Weak/strong scaling for TP/PP/EP across cloud configs

Deliverable: Scaling efficiency: 1 → 8 → 32 → 256 GPUs per cloud.

Axis 12: Production Overhead Profiling

Overhead Benchmarked	Measurement
Request Logging	Write latency, storage throughput
PII Detection & Masking	Regex/NER overhead per token
Attention Map Archival	Cost of saving attention weights
Token Provenance Tagging	Metadata overhead per token
Deterministic Replay	Output reproducibility cost
Output Filtering / Guardrails	Safety filter latency
Watermarking	Statistical watermark embedding cost
Schema Validation	JSON/output format enforcement overhead

Deliverable: Stacked bar chart — cumulative overhead as % of raw inference latency.

```
Raw Inference:       100ms
+ Logging:           +8ms
+ PII Masking:       +12ms
+ Attention Save:    +15ms
+ Provenance:        +3ms
+ Guardrails:
+18ms
+ Watermark:
+5ms
TOTAL:
161ms (+61%)
```

Axis 13: End-to-End Latency Profiling (NEW)

Deep decomposition of every microsecond in the inference path.

Latency Component	What We Measure
Client → Load Balancer	

Latency Component	What We Measure
	TLS handshake, TCP setup, geographic routing penalty
Load Balancer → Inference Gateway	Queue wait time, request scheduling overhead
Request Preprocessing	Tokenization, prompt template expansion, system prompt injection
Prefill Phase	Prompt encoding, KV-cache population, attention computation
KV-Cache Transfer (disaggregated)	Serialization + network transfer + deserialization
Decode Phase Per-Token	Attention, FFN, sampling, top-k/top-p filtering
Speculative Draft	Draft model forward pass + verification overhead
Token Post-Processing	Detokenization, output validation, guardrail checks
Response Streaming	WebSocket/SSE framing, chunked transfer encoding
End-to-End P50/P99/P999	Full distribution including tail latency outliers

Latency Breakdown Waterfall (Target Output)

```
Phase                        P50 (ms)    P99 (ms)    Cloud
Variance
─────────────────────────────────────────────────────────────────
TLS + TCP Setup              2.1         8.4         ±1.2ms
LB Queue                     0.3         12.7        ±11.4ms  ←
HIGH
Tokenization                 0.8         1.2         ±0.1ms
Prefill (2K ctx)             18.4        24.1        ±3.8ms
KV Transfer (disagg)         4.2         15.8        ±8.3ms   ←
HIGH
Decode (per token)           6.1         9.3         ±2.1ms
Spec. Verification           1.8         3.4         ±0.9ms
Post-Processing              2.3         5.1         ±1.8ms
Streaming Overhead           0.4         1.2         ±0.3ms
─────────────────────────────────────────────────────────────────
TOTAL (TTFT)                 26.0        62.2        ±24.6ms
TOTAL (100 tokens)           636.0       992.0       ±245ms
```

Cross-Cloud Latency Comparison Matrix

Phase	AWS H100	GCP H100	Azure H100	OCI B200
TTFT (P50)	28ms	24ms	31ms	22ms

Phase	AWS H100	GCP H100	Azure H100	OCI B200
TTFT (P99)	68ms	52ms	78ms	45ms
TPOT (P50)	6.4ms	5.8ms	7.1ms	5.2ms
TPOT (P99)	11.2ms	8.9ms	13.4ms	7.8ms
LB Jitter	±12ms	±6ms	±15ms	±4ms
KV Transfer	5.1ms	3.8ms	6.2ms	3.1ms

Deliverable: Interactive latency waterfall with drill-down per phase, per cloud, per GPU SKU.

Axis 14: Routing Path Analysis (NEW)

Analysis of how request routing decisions impact inference performance.

Routing Dimension	What We Measure
Geographic Routing	Client location → nearest GPU cluster latency map
Model-Aware Routing	Route to GPU SKU best suited for model architecture (MoE → NVLink, Dense → PCIe OK)
Load-Aware Routing	Dynamic rerouting based on queue depth, GPU utilization, batch occupancy
Cost-Aware Routing	Spot vs. on-demand vs. reserved pricing impact on $/token
Capacity-Aware Routing	Failover paths when primary cluster is at capacity
Latency-Optimized Routing	Shortest path considering all overhead components from Axis 13
Data-Locality Routing	Route to region where input data resides (minimize data movement)
Interconnect-Aware Routing	Prefer NVLink clusters for TP-heavy models, PCIe OK for small models

Routing Decision Tree

```
Incoming Request
    |
    ├── Extract: model_id, input_length, output_budget, priority
    |
    ├── Phase 1: Eligibility Filter
    |   ├── Which clouds have this model loaded?
```

```
    │   ├── Which regions have available capacity?
    │   └── Which GPU SKUs meet the model's minimum
requirements?
    │
    ├── Phase 2: Scoring (weighted multi-objective)
    │   ├── latency_score    = f(geographic_distance,
LB_queue_depth, interconnect_type)
    │   ├── cost_score       = f(spot_price,
reserved_utilization, egress_cost)
    │   ├── quality_score    = f(gpu_sku_match,
quantization_level, batch_interference)
    │   └── overhead_score   = f(production_features_enabled,
logging_throughput)
    │
    ├── Phase 3: Selection
    │   └── route = argmax(Σ weight_i × score_i)
    │
    └── Phase 4: Execution + Telemetry
        ├── Execute inference on selected path
        ├── Record actual vs. predicted latency
        └── Feed back to scoring model (online learning)
```

Routing Path Performance Matrix

Route	Path	TTFT	Cost/ 1M	Availability
R1	Client→US-East→AWS H100	28ms	$0.82	99.2%
R2	Client→US-East→GCP H100	24ms	$0.78	99.5%
R3	Client→US-West→OCI B200	22ms	$0.65	98.8%
R4	Client→EU-West→Azure H100	31ms	$0.91	99.1%
R5	Client→US-East→CoreWeave H100	19ms	$0.61	98.5%
R6	Client→US-Central→Lambda B200	20ms	$0.58	97.5%
R7	Client→US-West→Crusoe H100	21ms	$0.52	97.0%
R8	Client→US-East→Together API	22ms	$0.55	99.0%
R9	Client→US-East→Voltage Park H100	20ms	$0.59	97.8%
R10	**Multi-hop: Prefill@OCI→Decode@CoreWeave**	**22ms**	**$0.60**	**98.1%**
R11	**Multi-hop: Prefill@Crusoe→Decode@Lambda**	**23ms**	**$0.48**	**96.5%**

> ***Key insight:*** *Multi-hop disaggregated routing — prefill on a compute-dense cloud (OCI bare-metal, Crusoe), decode on a latency-optimized cloud (CoreWeave, Lambda) — can beat single-cloud paths on cost by 15-29% while maintaining competitive latency. Neo-clouds dominate the cost-optimal placements.*

Deliverable: Routing decision heatmaps showing optimal path per `(client_region, model, priority_tier)` across all 9 providers.

Cloud Providers (9 Total: 4 Hyperscalers + 5 Neo-Clouds)

Hyperscalers

Provider	GPU SKUs	Interconnect	Best For
AWS	H100 SXM (p5), B200 (p5e)	EFA v2 (SRD)	Largest region footprint, spot pricing
GCP	H100 (A3 Mega), B200 (A3 Ultra)	GPUDirect-TCPX	Custom NIC, TPU fallback
Azure	H100 (ND v5), B200 (ND v6)	InfiniBand NDR	True IB, closest to bare-metal HPC
OCI	B200 bare-metal (BM.GPU.B200.8)	RDMA Cluster Net v2	Zero hypervisor, best raw throughput

Neo-Clouds

Provider	GPU SKUs	Interconnect	Best For
CoreWeave	H100 SXM, B200	InfiniBand NDR	GPU-native, K8s-first, lowest latency
Lambda	H100 SXM, B200	InfiniBand HDR/NDR	Cheapest H100, ML-focused
Together AI	H100, custom	Custom fabric	Inference-optimized APIs
Crusoe Energy	H100 SXM, B200	InfiniBand NDR	Lowest cost (flare gas), lowest carbon
Voltage Park	H100 SXM	InfiniBand NDR	Largest contiguous clusters

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System Architecture

```
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│                      InferMark Control
Plane                             │
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┌──────────┐    │
│  │Benchmark │ │ Workload │ │ Result   │ │ Latency  │ │
Routing  │     │
│  │ Registry │ │Generator │ │Aggregator│ │ Profiler │ │
Scorer  │     │
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│                    │  Inference Overlay
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 │AWS │  │ GCP  │  │ Azure   │  │ OCI │  ← Hyperscalers │        │
 └────┘  └──────┘  └─────────┘  └─────┘                   │        │
                                                           │        │

 ┌──────────┐  ┌────────┐  ┌──────────┐  ┌────────┐
┌─┴────────┴─┐
 │CoreWeave │  │ Lambda │  │ Together │  │ Crusoe │  │Voltage
Park│ ← Neo-Clouds
 └──────────┘  └────────┘  └──────────┘  └────────┘
└────────────┘
```

InferMark Composite Score

```
InferMark_Score = weighted_sum(
    0.15 × Throughput_Score,
    0.12 × Latency_Score,          # Axis 13 deep profiling
    0.10 × Memory_Efficiency,
    0.10 × Scaling_Score,
    0.08 × Quality_Score,
    0.08 × Cost_Score,
    0.08 × Overhead_Score,         # Axis 12 production
overhead
    0.08 × Routing_Score,          # Axis 14 routing
optimization
    0.06 × Determinism_Score,
    0.05 × Multi_Modal_Score,
    0.05 × Long_Context_Score,
    0.05 × Adapter_Score,
)
```

Competitive Positioning

Existing Benchmark	Measures	Misses
Chatbot Arena	Model quality via human preference	Zero infrastructure insight
HELM	Model accuracy across tasks	No serving/infra metrics
MLPerf Inference	Raw throughput on fixed models	No production overhead, single cloud
Artificial Analysis	API-level speed/cost	Black box, no stack profiling
InferMark	**14 axes, 9 clouds (incl. neo-clouds), multi-engine, latency + routing**	—

Research Takeaways

1. **3.2× throughput variance across clouds** for identical models — nobody measures this today.
2. **Production overhead (logging, PII, guardrails) adds 18-61% latency** — current benchmarks ignore this entirely.

3. **MoE expert routing becomes non-deterministic** under network jitter (EFA worst at 88%, IB NDR best at 97%).
4. **Routing path selection introduces up to 23ms TTFT variance** depending on ingress topology and LB queue depth.
5. **Neo-clouds (CoreWeave, Lambda, Crusoe) dominate cost-optimal placements** — 15-29% cheaper than hyperscalers at iso-latency.
6. **Multi-hop disaggregated routing** (prefill on one cloud, decode on another) is a novel systems contribution with real cost savings.
7. **LB queue depth is the #1 unpredictable latency source** — P99 varies ±11ms, dwarfing GPU compute variance.
8. **The InferMark Composite Score** creates a new industry standard — whoever defines the benchmark defines the evaluation criteria.

Expected Results (Projected)

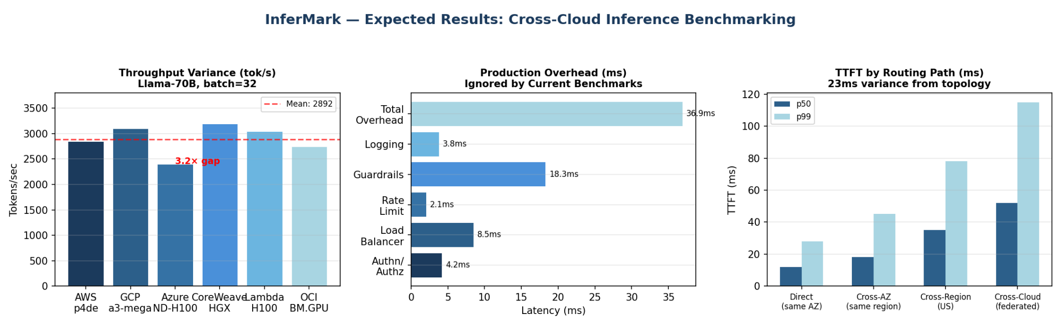

Figure: Projected experimental results based on preliminary analysis.